

COOCH BEHAR GOVERNMENT ENGINEERING COLLEGE

Department of Computer Science & Engineering

ASSIGNMENT-II [CSE 3RD SEMESTER]

IT WORKSHOP LAB [Python Programming]

1. Write a recursive program to solve The Tower of Hanoi Problem.
2. Write a recursive function to calculate the sum of digits of the n-digit number.
3. Write a recursive function to find the binary equivalent of a positive integer.
4. Write a recursive program without Tail recursion to calculate sum of first n natural numbers.
5. Write the function which removes all the vowels in a word and returns the remaining letters in the word.
6. Write a program to convert a hexadecimal number entered as a string into its equivalent binary number.
7. Write a function check_duplicate(L) which returns True if a list L contains duplicates elements . It should return False if all the elements in L are unique.
8. Write a program to implement linear search.
9. Write a program to implement binary search.
10. Write a program to implement Insertion sort.
11. Write a program to implement merge sort.
12. Write a program to implement quick sort.
13. Write a program to perform the following operations on complex numbers **using class**.
 - a) Addition
 - b) Subtraction
 - c) Multiplication
 - d) Check two complex numbers are equal or not
14. Write a program to copy the contents of one file to another.
15. Write a program to read the content of a python file and display all the lines without comments.